Linear systems and elimination

Exploit spectral outliers without losing input sparsity

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Topic: sparse linear systems; randomized preconditioning

Rating rationale: Challenging reflects combining spectral deflation with an input-sparsity cost guarantee; community impact spans sparse solvers, preconditioning, and regression.

Partial results — Sidney Holden, 12 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) gives a deterministic $O(N \min\{n, k+1+\kappa \log(2/\varepsilon)\})$ solver in Theorem 2.1. Corollary 2.2 achieves this entry's target when $k = 0$, $k \le \kappa$, $N \le k^{\omega_0 - 1}$, or $n \le 2k$.

Theorem 4.1 establishes the additive sparsity/outlier bound for explicitly sparse SPD matrices with an **exactly equal eigenvalue tail**. Corollary 5.1 covers an exactly flat singular tail with cost depending on the sum of squared row supports; that quantity can be quadratic in input sparsity. The counterexamples in Section 3 invalidate particular shortcuts, not all possible algorithms.

The general bounded-tail sparse-input target below remains open. The restricted results passed a separate independent informal Codex AI-agent audit. This is not external human peer review or formal verification; no Lean verification was performed. Source, attribution, verified affiliation and reproducibility record.

Context and notation

All norms are Euclidean vector norms or their induced matrix norms.

Problem statement

Let $A \in \mathbb{R}^{n \times n}$ be nonsingular, with singular values $\sigma_1 \ge \cdots \ge \sigma_n > 0$, and write $N = \mathrm{nnz}(A)$ and $\kappa_2(A) = \sigma_1/\sigma_n$. The input consists of its nonzero entries and their indices, $b \ne 0$, $\varepsilon \in (0, 1/2)$, $k \in \{0, \ldots, n-1\}$, and $\kappa \ge 1$, with the promise $\sigma_{k+1}/\sigma_n \le \kappa$. Fix any exponent $\omega_0 > 2$ for which square matrix multiplication has an $O(m^{\omega_0})$ arithmetic algorithm.

Does a randomized algorithm exist that, for every such input, produces $\hat{x}$ satisfying

$$\|A\hat{x} - b\|_2 \le \varepsilon \|b\|_2$$

with probability at least 0.99, using at most

$$C(k^{\omega_0} + N\kappa)[1 + \log(n\kappa_2(A)/\varepsilon)]^q$$

exact arithmetic operations? Here C, q may depend on the chosen multiplication algorithm, but not on the input. Direct entry access is allowed. The display makes the workshop's suppressed logarithms and accuracy convention explicit; ω_0 avoids assuming that the infimum defining the multiplication exponent is attained.

References

Amsel et al., *Linear Systems and Eigenvalue Problems*, Definition 2.4 and Problem 2.5. Dereziński and Sidford, *Approaching Optimality for Solving Dense Linear Systems with Low-Rank Structure*, SODA 2026, pp. 925–938; full manuscript, Theorems 3 and 29, gives the dense-input predecessor.

Earlier status check — 2026-09-08

Searches for *sparse outlying singular values solver 2026* and *spectral outliers linear systems 2026* found the newer Liu, Nguyen, Peng, and Yang *Faster Solvers for Sparse Systems with Large Spectral Outliers*. Its Theorem 1.1 assumes an explicitly given factor B in $B^T Bx = b$, bounded spectral tail, polynomial conditioning, and short rational input. For an $m \times d$ factor with at most s nonzeros per row, its bit bound includes $m^{o(1)}\widetilde{O}(ms + \sqrt{d/k}\,k^2 + k^{\omega+\eta})$. These input restrictions and extra terms do not establish the requested general bound. The workshop's August update records this as partial progress on Problem 2.9.

Audit update — 2026-09-10

Rechecked workshop Problem 2.5 and Liu–Nguyen–Peng–Yang, Theorem 1.1. The latter is a meaningful newer sparse-factor result but retains different input assumptions and overhead, as detailed above. Spectral-outlier solver searches found no theorem establishing this entry's exact general bound; the author-hosted manuscript is identifiable through Liu's publication list.